Applied Linear Algebra

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TAs

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This week

- Topics to be covered
  - Vectors
  - Norm and Distance
  - Matrices
  - Systems of Linear Equations
  - Least Squares Fitting

- Applications
  - Linear Algebra with Python
  - Linear Algebra in the Real World

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Chapter 1: Vectors

1. Notations
2. Examples
3. Addition and Scalar Multiplication
4. The dot product

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Vectors

- A vector is an ordered list of numbers written as

$$\begin{bmatrix} -1.1 \\ 2.1 \\ 0.0 \\ -4.5 \end{bmatrix} \quad \text{or} \quad \begin{pmatrix} -1.1 \\ 2.1 \\ 0.0 \\ -4.5 \end{pmatrix} \quad \text{or} \quad (-1.1, 2.1, 0.0, -4.5)$$

- The numbers in the list are the *entries or components*

- The number of elements is the *size (dimension)* of the vector

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Vector Notations

- We often use symbols to denote vectors, e.g., $\mathbf{x}, \mathbf{a}, \mathbf{u}, \mathbf{p}, \beta, \mathbf{X}$..
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- *Note:* sometimes $\mathbf{u_i}$ refers to the ith vector in a list of vectors

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Zeros, Ones, and Unit Vectors

- An n-vector with all entries $\mathbf{0}$ is denoted $\mathbf{0_n}$ or just $\mathbf{0}$.
- An vector of size n with all entries $\mathbf{1}$ is denoted $\mathbf{1_n}$ or just $\mathbf{1}$.
- A standard unit vector $\mathbf{e_i}$ has one entry 1 in the i position and all others 0

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Sparse Vectors

- A vector is said to be *sparse* if many of its entries are zero,
- It is stored efficiently on the computer
- $\mathbf{nnz}(x)$ is number of entries that are nonzero

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Example: Location and Displacement in 2D or $\mathbb{R}^2$:

- A two vector can be used to represent a position or a location in a 2D space (same as in a 3D space)

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Example: Color:

- All colors on the RGB color model can also be created by layering all three primary colors on top of each other at different brightnesses. The intensity values are represented by 0 and 1

- : $(1, 0, 0)$: $(0, 1, 0)$: $(0, 0, 1)$.

- Mix Red and Green to get : $(1, 1, 0)$

- Mix RGB to get : $(0.5, 0.5, 0.5)$

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More Examples

- Values across a population: an n-vector $\mathbf{b}$ gives some quantities (for ex: blood pressure) for person i.

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More Examples

- Values across a population: an n-vector $\mathbf{b}$ gives some quantities (for ex: blood pressure) for person i.
- Proportion: a vector $\mathbf{w}$ that give fractions out of n choices.
- Features and Attributes: x_i is the value of ith feature or attribute of an entity
  asset

Equal Vectors and Vector Addition

- two vectors $\mathbf{u}$ and $\mathbf{v}$ of the **same size** are equal if $u_i = v_i$ for all i. We write this as $\mathbf{u} = \mathbf{v}$

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- $\mathbf{u}$ and $\mathbf{v}$ of the **same size** can be added, and sum denoted $\mathbf{u} + \mathbf{v}$
- to get the sum, add corresponding entries
- subtraction is similar

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Properties of vector addition

- commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- associative: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
- $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
- $\mathbf{u} - \mathbf{u} = \mathbf{0}$

Example: Adding Displacements:

- If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in 2D or 3D, then $\mathbf{u} + \mathbf{v}$ is the sum displacement

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Displacements between two points:

- If $\mathbf{p}$ and $\mathbf{q}$ represent two the positions of two points in 2D or 3D then $\mathbf{p} - \mathbf{q}$ is the displacement vector from $\mathbf{q}$ to $\mathbf{p}$

Scalar-vector multiplication:

- Scalar c and n-vector $\mathbf{u}$ can be multiplied:

$$c\mathbf{u} = (cu_1, cu_2, \ldots cu_n)$$

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Properties of scalar-vector multiplication

- associative: $(cd)\mathbf{u} = c(d\mathbf{u})$
- distributive: $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
- distributive: $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$

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Linear Combinations

- for vectors $\mathbf{u_1}, \ldots, \mathbf{u_m}$ and scalars $c_1, \ldots, c_m$,

$$c_1\mathbf{u_1} + \ldots + c_m\mathbf{u_m}$$

  is a linear combination of the vectors $c_1, \ldots, c_m$.

- the scalars $c_1, \ldots, c_m$ are called the coefficients of the linear combinations.

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Example: Displacements

Example: What are all combinations $c\mathbf{u}$?

Example: What are all combinations $c\mathbf{u} + d\mathbf{v}$?

Dot Product

- $$\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$$

$$\boxed{\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2}$$

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- The dot product of two vectors **u** and **v** is

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- Other notation used: $\mathbf{u} \cdot \mathbf{v}, \langle \mathbf{u}, \mathbf{v} \rangle, \langle \mathbf{u} \mid \mathbf{v} \rangle$

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Properties of inner product

- $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{c}(\mathbf{u} \cdot \mathbf{v})$
- $(\mathbf{u} + \mathbf{v}) \cdot w = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

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Applications

- $\mathbf{p}$ is vector of prices, $\mathbf{q}$ is vector of quantities; $\mathbf{p} \cdot \mathbf{q}$ is total cost

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- $\mathbf{w}$ is weight vector, $\mathbf{f}$ is feature vector; $\mathbf{w} \cdot \mathbf{f}$ is score